

ECON 0100 | Vignette B6 | International Trade

I Solutions

Answers and work are shown in the red blocks.

Members of the wizarding world have preferences for pumpkin pasties according to the following demand curve:

$$P_d = 14 - \frac{1}{2}Q_d$$

Pumpkin pasties are produced by many sellers according to the following supply curve:

$$P = 2 + \frac{1}{2}Q_s$$

Prices are in galleons and quantity is in pasties.

Q1 | {++The UK's Pumpkin Pasties++}

Economic historians remember a time before the Ministry lifted barriers to exports and imports of {++pumpkin pasties++} ~~lemon tarts~~. The global price was 6 galleons. However, due to the trade barriers the UK did not ~~did not~~ have access to this market for {++pasties++} ~~tarts~~. {++The UK's domestic supply and demand curves are the ones above.++}

Use a graph and algebra to illustrate what happened to the market after the trade barrier was lifted.

- a) Imports or exports? **imports**, exports
- b) Quantity demanded: **16** Quantity supplied: **8** Imports: **8**
- c) Consumer surplus before: **36** after: **64**
- d) Producer surplus before: **36** after: **16**
- e) Change in total surplus: **+8**

f) Buyers: **gain**, lose Sellers: gain, **lose**

Closed market: 12 pasties at 8 galleons, $CS = PS = 36$.

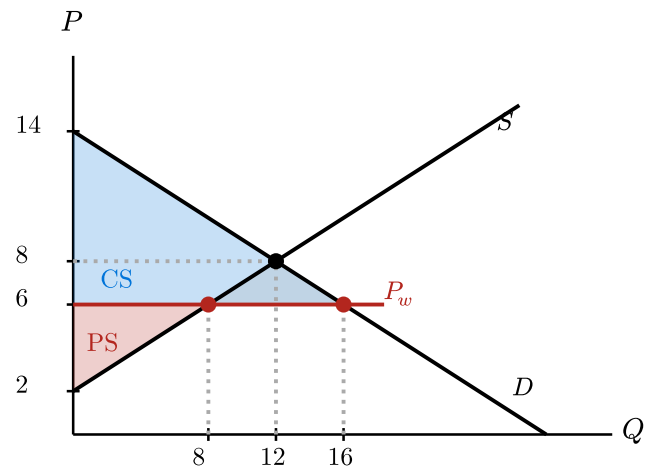
At the world price of 6, below the domestic price, buyers buy abroad: the market **imports**.

Quantity demanded 16, quantity supplied 8; imports $16 - 8 = 8$.

$CS = \frac{1}{2} \cdot 16 \cdot (14 - 6) = 64 (+28)$. $PS = \frac{1}{2} \cdot 8 \cdot (6 - 2) = 16 (-20)$.

Total surplus rises by 8, the triangle $\frac{1}{2} \cdot 8 \cdot |8 - 6|$ between the curves and the world price.

Buyers gain more than sellers lose.



Q2 | {++Egypt's Pumpkin Pasties++}

While the supply curve is very similar to that of the UK, demand for {++pumpkin pasties++} ~~lemon tarts~~ is much lower in Egypt's wizarding community, leading to a domestic price below the global price. Use a graph (*but no algebra*) to illustrate what happened to the market after Egypt lifted their ban on international trade of {++pumpkin pasties++} ~~lemon tarts~~.

a) Egypt imports, **exports**

b) Buyers: gain, **lose** Sellers: **gain**, lose

c) Total surplus: **rises**, stays the same, falls

With low demand, Egypt's domestic price is below the world price, so sellers sell abroad at the world price and Egypt **exports**. Buyers now pay the higher world price and lose; sellers gain more than buyers lose, so total surplus rises by the triangle between the curves and the world price.

Q3 | {++A Rising World Price++}

{++Back in the UK, suppose the global price rises from 6 to 7 galleons.++}

a) Quantity demanded: **14** Quantity supplied: **10** Imports: **4**

b) Imports have: risen, stayed the same, **fallen**

At 7, buyers want 14 and domestic sellers supply 10, so imports fall from 8 to **4**. A higher world price moves the market back toward the closed-market equilibrium: CS falls to 49, PS rises to 25, and the gain from trade shrinks to 2.